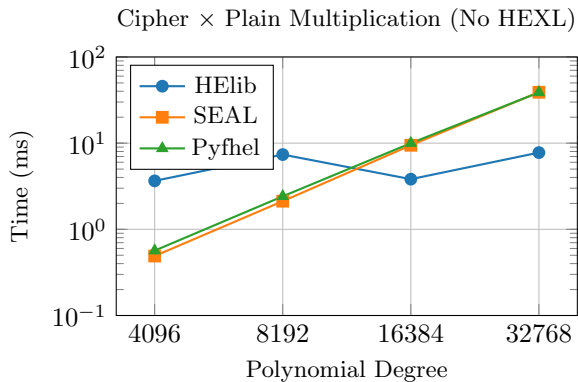
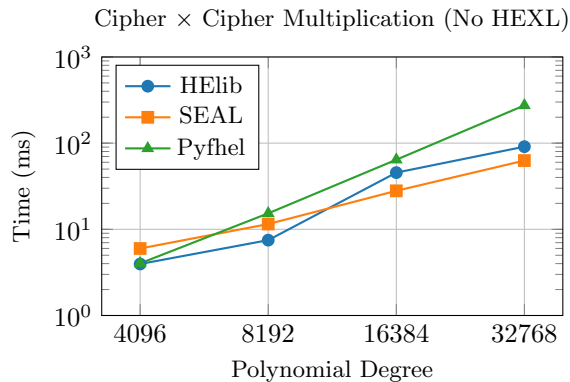
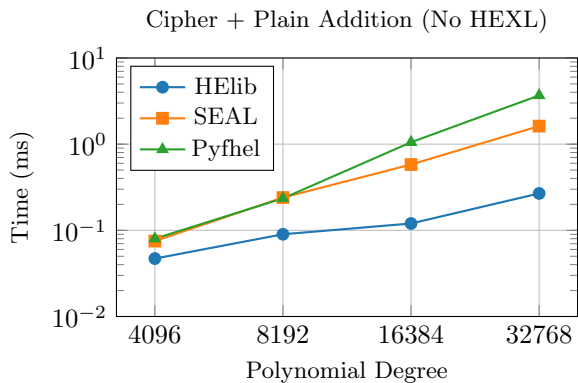
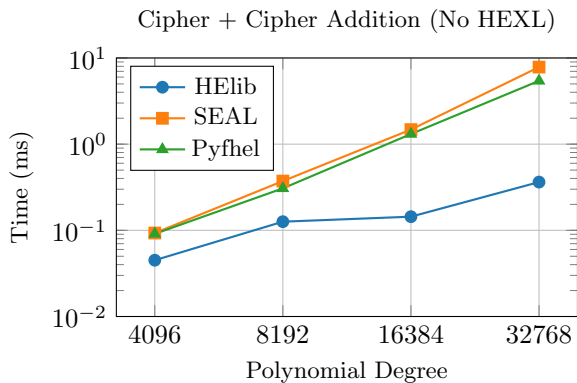




Performance comparison of individual FHE operations across HELib, SEAL, and Pyfhel libraries without HEXL acceleration. **Top-left:** Ciphertext-ciphertext addition shows HELib's dominance. **Top-right:** Ciphertext-plaintext addition demonstrates HELib's consistent advantage. **Bottom-left:** Ciphertext-ciphertext multiplication reveals SEAL's efficiency at higher polynomial degrees. **Bottom-right:** Ciphertext-plaintext multiplication shows SEAL's strong performance across all polynomial degrees.